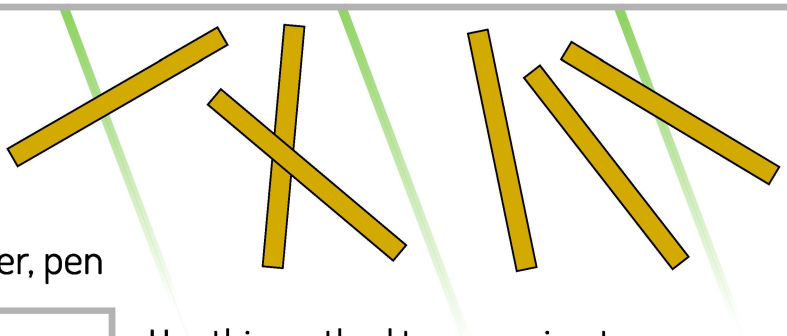
MathsJam Shout

March 2025
π Day Special

Try It Buffon's Needle



You will need: matchsticks, paper, ruler, pen

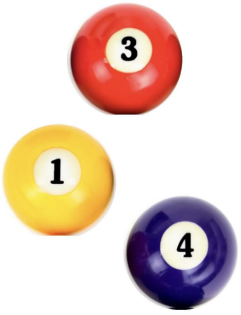
Use this method to approximate π:
Draw parallel lines on a piece of paper, and drop matchsticks all the same length (or the same one multiple times) randomly on to it. Count how many of the matchsticks cross a line, and how many don't. The ratio, along with the relative length of matchsticks and distance between lines, will allow you to calculate an approximation for π.

- Can you prove why this works?
- What about on a 2D grid of lines? Search for 'Buffon-Laplace Needle Problem'.
- Search for 'Buffon's Noodle' for a twist on the problem!

Instructions and calculations here::
bit.ly/buffon-method

What Finding π on a pool table

Improbably, the value of π can be found in the number of bounces a pool ball makes between the edge of a pool table and another incoming ball before it stops. If your pub has a pool table, why not try it for yourself?

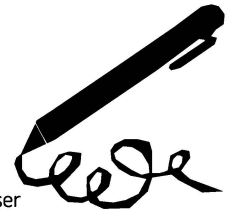


Watch Matt Parker's latest YouTube video, where he tries it for real:
youtube.com/standupmaths
Read this Aperiodical post by Colin Wright:
bit.ly/pi-billiards

Write A π Poem

Use the digits of π to determine the length of each word in a poem, or the number of words on each line. Or write a haiku (π-ku) with 3, 1 and 4 syllables on each line. Share your poetry with other Jams!

Example π-ku:
Rational?
π?
Mumbo jumbo!
Colin Beveridge
Weymouth MathsJam organiser



MathsJam Shout is a monthly sheet of ideas for activities to do at a MathsJam night. It's created using suggestions from a different MathsJam each month, and if you'd like to submit suggestions for a month in the future, email **katie@mathsjam.com** for details.

MathsJam is a monthly opportunity for like-minded self-confessed maths enthusiasts to get together in a pub and share stuff they like. Puzzles, games, problems, or just anything they think is cool or interesting. Monthly MathsJam nights happen in over 70 locations around the world, on the second-to-last Tuesday of each month. To find your nearest MathsJam, visit the website at **www.mathsjam.com**.

INSTRUCTIONS

Make the following graph on a piece of paper to write your information.

TOOTHPICK	TALLY	TOTAL
Crosses Line		
Not Touching Line		

1. Measure your toothpick in centimeters. The length of your toothpick will determine how far apart your lines are on the paper.
2. On a piece of paper, starting at the top, draw parallel lines across the paper until you fill the page. The distance between the lines should be equal to the length of the toothpick.
3. Place the piece of paper on a flat surface. Drop the toothpick onto the paper from a height of 10 to 15 cm. Record where it lands.
4. Repeat Step 4 several times. The more times you drop the toothpick, and record the results, the more accurate your results.
5. After you have finished step 5, solve the following problem: (number of tosses)/(number of intersections).
6. Take your answer and multiply by 2. This is your approximation of π. Your answer should be close to 3.14.
7. Complete extension activities.

EXTENSIONS

- Try this experiment with different shapes with a computer simulation at: *https://mste.illinois.edu/activity/buffon/*
- Use Integral geometry to write a proof for the Buffon needle: *https://medium.com/however-mathematics/a-beautiful-way-to-calculate-%CF%80-buffons-needle-problem-51d41029f287*

RESOURCES

- https://mste.illinois.edu/activity/buffon/
- https://www.mathsisfun.com/activity/buffons-needle.html
- https://medium.com/however-mathematics/a-beautiful-way-to-calculate-%CF%80-buffons-needle-problem-51d41029f287

LEARN MORE

Students + Parents + Educators

For information on grants, training and student opportunities, curriculum ideas, and resources, please visit ***stem.inl.gov.***